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DBMS LAB ASSIGNMENT-1

MySQL – Questions, SQL Queries and Expected Outputs

QUESTION 1. Display all data from the existing default Employees (EMP or Employee) table.

If your table is named EMP or Employee, replace employees with that exact table name.

SQL QUERY:

```
SELECT * FROM employees;
```

EXPECTED OUTPUT:

```
The query displays all rows and columns in the existing employees table. The exact records depend on the database.
```

QUESTION 2. Display the structure of the Employees table.

Use the same table name as in Question 1.

SQL QUERY:

```
DESC employees;
```

EXPECTED OUTPUT:

```
The result displays column names, data types, NULL settings, key information and default values.
```

QUESTION 3. Create a table named Student with the following structure.

SQL QUERY:

```
CREATE TABLE Student (
    Roll DECIMAL(5,0),
    Name VARCHAR(30),
    Age DECIMAL(5,0),
    Course VARCHAR(5),
    Math DECIMAL(6,2),
    Physics DECIMAL(6,2),
    Computer DECIMAL(6,2),
    Birthday DATE
);
```

EXPECTED OUTPUT:

```
Query OK, 0 rows affected

Table structure:
Field       Data type
Roll        DECIMAL(5,0)
Name        VARCHAR(30)
Age         DECIMAL(5,0)
Course      VARCHAR(5)
Math        DECIMAL(6,2)
Physics     DECIMAL(6,2)
Computer    DECIMAL(6,2)
Birthday    DATE
```

QUESTION 4. Create the MSc table from the Student table with the same fields and structure but without any data.

SQL QUERY:

```
CREATE TABLE MSc LIKE Student;
```

EXPECTED OUTPUT:

```
Query OK, 0 rows affected
```

QUESTION 5. Display the structure of the MSc table.

SQL QUERY:

```
DESC MSc;
```

EXPECTED OUTPUT:

```
Field       Data type
Roll        DECIMAL(5,0)
Name        VARCHAR(30)
Age         DECIMAL(5,0)
Course      VARCHAR(5)
Math        DECIMAL(6,2)
Physics     DECIMAL(6,2)
Computer    DECIMAL(6,2)
Birthday    DATE
```

QUESTION 6. Create the MCA table from the Student table without data. Rename Course as Department and Name as First_Name.

SQL QUERY:

```
CREATE TABLE MCA LIKE Student;

ALTER TABLE MCA
CHANGE Course Department VARCHAR(5);

ALTER TABLE MCA
CHANGE Name First_Name VARCHAR(30);
```

EXPECTED OUTPUT:

```
Query OK. The MCA table is created with the renamed columns.
```

QUESTION 7. Display the structure of the MCA table.

SQL QUERY:

```
DESC MCA;
```

EXPECTED OUTPUT:

```
Field       Data type
Roll        DECIMAL(5,0)
First_Name  VARCHAR(30)
Age         DECIMAL(5,0)
Department  VARCHAR(5)
Math        DECIMAL(6,2)
Physics     DECIMAL(6,2)
Computer    DECIMAL(6,2)
Birthday    DATE
```

QUESTION 8. Insert the following records into the Student table.

SQL QUERY:

```
INSERT INTO Student VALUES
(1,'Rahul',19,'BCA',79.5,67,89,'1993-06-15'),
(2,'Kunal',21,'BCA',68,76,59.5,'1991-08-16'),
(3,'Aditi',20,'MSc',90,73,56,'1992-09-20'),
(4,'Sumit',20,'MCA',57.5,78,81,'1991-12-07'),
(5,'Anirban',22,'MCA',80,68,63,'1994-09-15'),
(6,'Kumkum',21,'BCA',72,54.5,60,'1995-02-08'),
(7,'Suman',21,'BCA',91.5,32,61,'1994-03-10'),
(8,'Rohit',22,'MSc',85,76,92,'1992-04-19');
```

EXPECTED OUTPUT:

```
Query OK, 8 rows affected

Roll  Name     Age  Course  Math   Physics  Computer  Birthday
1     Rahul    19   BCA     79.50  67.00    89.00     1993-06-15
2     Kunal    21   BCA     68.00  76.00    59.50     1991-08-16
3     Aditi    20   MSc     90.00  73.00    56.00     1992-09-20
4     Sumit    20   MCA     57.50  78.00    81.00     1991-12-07
5     Anirban  22   MCA     80.00  68.00    63.00     1994-09-15
6     Kumkum   21   BCA     72.00  54.50    60.00     1995-02-08
7     Suman    21   BCA     91.50  32.00    61.00     1994-03-10
8     Rohit    22   MSc     85.00  76.00    92.00     1992-04-19
```

QUESTION 9. Display all students' details from the Student table.

SQL QUERY:

```
SELECT * FROM Student;
```

EXPECTED OUTPUT:

```
Roll  Name     Age  Course  Math   Physics  Computer  Birthday
1     Rahul    19   BCA     79.50  67.00    89.00     1993-06-15
```

```
2     Kunal    21   BCA     68.00  76.00    59.50     1991-08-16
3     Aditi    20   MSc     90.00  73.00    56.00     1992-09-20
4     Sumit    20   MCA     57.50  78.00    81.00     1991-12-07
5     Anirban  22   MCA     80.00  68.00    63.00     1994-09-15
6     Kumkum   21   BCA     72.00  54.50    60.00     1995-02-08
7     Suman    21   BCA     91.50  32.00    61.00     1994-03-10
8     Rohit    22   MSc     85.00  76.00    92.00     1992-04-19
```

QUESTION 10. Find the details of the student with Roll No. 5 from the Student table.

SQL QUERY:

```
SELECT * FROM Student
WHERE Roll = 5;
```

EXPECTED OUTPUT:

```
Roll  Name     Age  Course  Math   Physics  Computer  Birthday
5     Anirban  22   MCA     80.00  68.00    63.00     1994-09-15
```

QUESTION 11. Show the Roll, Name and marks of all subjects for all students from the Student table.

SQL QUERY:

```
SELECT Roll, Name, Math, Physics, Computer
FROM Student;
```

EXPECTED OUTPUT:

```
Roll  Name     Math   Physics  Computer
1     Rahul    79.50  67.00    89.00
2     Kunal    68.00  76.00    59.50
3     Aditi    90.00  73.00    56.00
4     Sumit    57.50  78.00    81.00
5     Anirban  80.00  68.00    63.00
6     Kumkum   72.00  54.50    60.00
7     Suman    91.50  32.00    61.00
8     Rohit    85.00  76.00    92.00
```

QUESTION 12. Display all student details belonging to the BCA department.

SQL QUERY:

```
SELECT * FROM Student
WHERE Course = 'BCA';
```

EXPECTED OUTPUT:

```
Roll  Name    Age  Course  Math   Physics  Computer  Birthday
1     Rahul   19   BCA     79.50  67.00    89.00     1993-06-15
2     Kunal   21   BCA     68.00  76.00    59.50     1991-08-16
6     Kumkum  21   BCA     72.00  54.50    60.00     1995-02-08
7     Suman   21   BCA     91.50  32.00    61.00     1994-03-10
```

QUESTION 13. Insert data into the MCA table from the Student table where Course is MCA.

SQL QUERY:

```
INSERT INTO MCA
(Roll, First_Name, Age, Department, Math, Physics, Computer, Birthday)
SELECT Roll, Name, Age, Course, Math, Physics, Computer, Birthday
FROM Student
WHERE Course = 'MCA';
```

EXPECTED OUTPUT:

```
Query OK, 2 rows affected
```

QUESTION 14. Insert data into the MSc table from the Student table where Course is MSc.

SQL QUERY:

```
INSERT INTO MSc
SELECT * FROM Student
WHERE Course = 'MSc';
```

EXPECTED OUTPUT:

```
Query OK, 2 rows affected
```

QUESTION 15. Display the structure of the Student and MCA tables.

SQL QUERY:

```
DESC Student;
DESC MCA;
```

EXPECTED OUTPUT:

```
Student contains Roll, Name, Age, Course, Math, Physics, Computer and Birthday. MCA contains Roll, First_Name, Age, Department
```

QUESTION 16. Display all details of each student in the Student table with Course appearing first in the display column list.

SQL QUERY:

```
SELECT Course, Roll, Name, Age, Math, Physics, Computer, Birthday
FROM Student;
```

EXPECTED OUTPUT:

```
Course  Roll  Name     Age  Math   Physics  Computer  Birthday
BCA      1     Rahul    19   79.50  67.00    89.00     1993-06-15
BCA      2     Kunal    21   68.00  76.00    59.50     1991-08-16
MSc      3     Aditi    20   90.00  73.00    56.00     1992-09-20
MCA      4     Sumit    20   57.50  78.00    81.00     1991-12-07
MCA      5     Anirban  22   80.00  68.00    63.00     1994-09-15
BCA      6     Kumkum   21   72.00  54.50    60.00     1995-02-08
BCA      7     Suman    21   91.50  32.00    61.00     1994-03-10
MSc      8     Rohit    22   85.00  76.00    92.00     1992-04-19
```

QUESTION 17. Update the Math marks of the student with Roll No. 7 from 91 to 95 in the Student table.

SQL QUERY:

```
UPDATE Student
SET Math = 95
WHERE Roll = 7;
```

EXPECTED OUTPUT:

```
Query OK, 1 row affected

Check:
SELECT Roll, Name, Math FROM Student WHERE Roll = 7;

Expected row: 7 | Suman | 95.00
```

QUESTION 18. Update the name of the student with Roll No. 4 from Sumit to Sumitava in the MCA table.

SQL QUERY:

```
UPDATE MCA
SET First_Name = 'Sumitava'
WHERE Roll = 4;
```

EXPECTED OUTPUT:

```
Query OK, 1 row affected

Check:
SELECT Roll, First_Name, Department FROM MCA;

Expected rows:
4 | Sumitava | MCA
5 | Anirban  | MCA
```

QUESTION 19. Delete the details of the student with Roll No. 2 from the Student table.

SQL QUERY:

```
DELETE FROM Student
WHERE Roll = 2;
```

EXPECTED OUTPUT:

```
Query OK, 1 row affected
```

QUESTION 20. Delete all student details from the Student table.

SQL QUERY:

```
DELETE FROM Student;
```

EXPECTED OUTPUT:

```
Query OK, 7 rows affected. This assumes Question 19 has already been executed, so seven records remain before this query.
```